END TERM EXAMINATION - COMPUTER SCIENCE (2023-24)

Time: 3 Hours **Max Marks: 100**

Instructions: Answer all questions in Part A.

PART A

Q1. What is Machine Learning? **[7 Marks]**

Machine learning is a subset of artificial intelligence (AI) that focuses on building systems that learn from data, identify patterns, and make decisions with minimal human intervention. Key types include Supervised Learning (using labeled data), Unsupervised Learning (finding patterns in unlabeled data), and Reinforcement Learning (learning through rewards and punishments). Real-world applications span image recognition, recommendation engines, and predictive maintenance.

Q2. What is Overfitting? **[5 Marks]**

Overfitting is a modeling error that occurs when a machine learning function fits too closely to the training data, capturing noise and random fluctuations rather than the underlying pattern. This leads to high accuracy on the training set but poor generalization to new, unseen data. It can be caused by overly complex models or insufficient training data.

Q3. What is Decision Trees? **[3 Marks]**

A decision tree is a non-parametric supervised learning method used for both classification and regression. It uses a tree-like model of decisions and their possible consequences, branching from a root node through internal nodes to leaf nodes that represent outcomes.